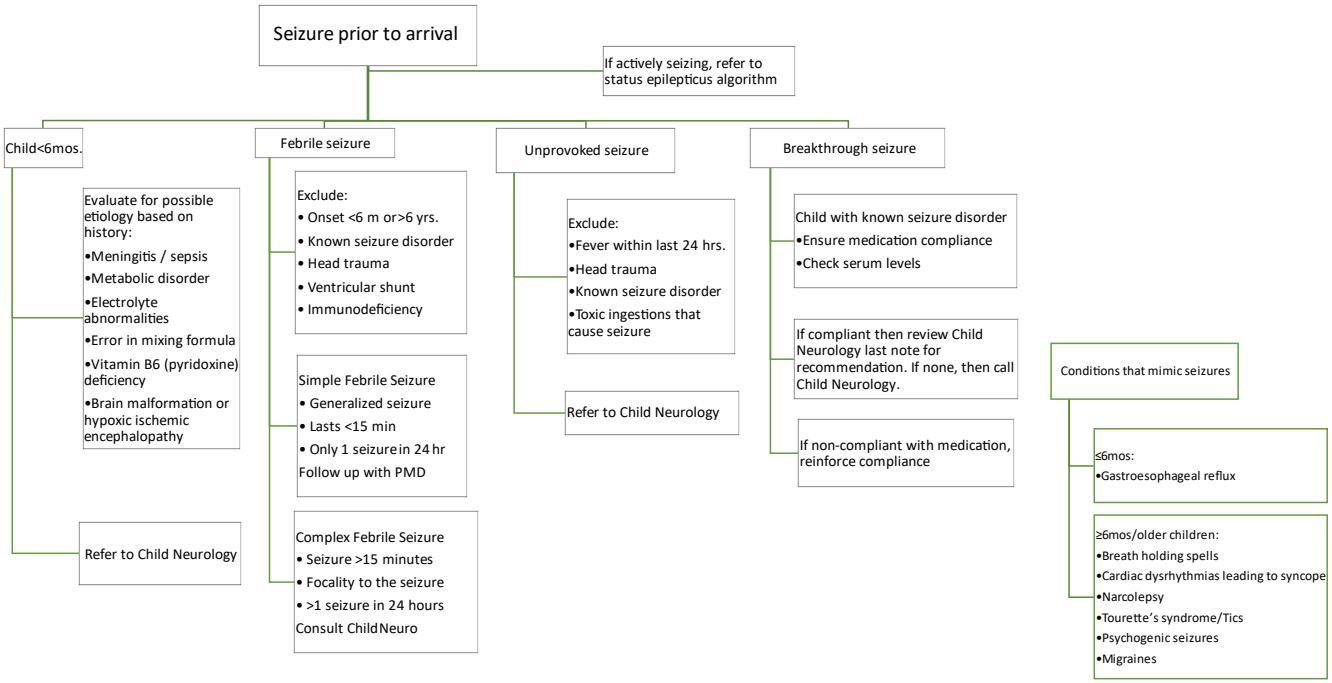
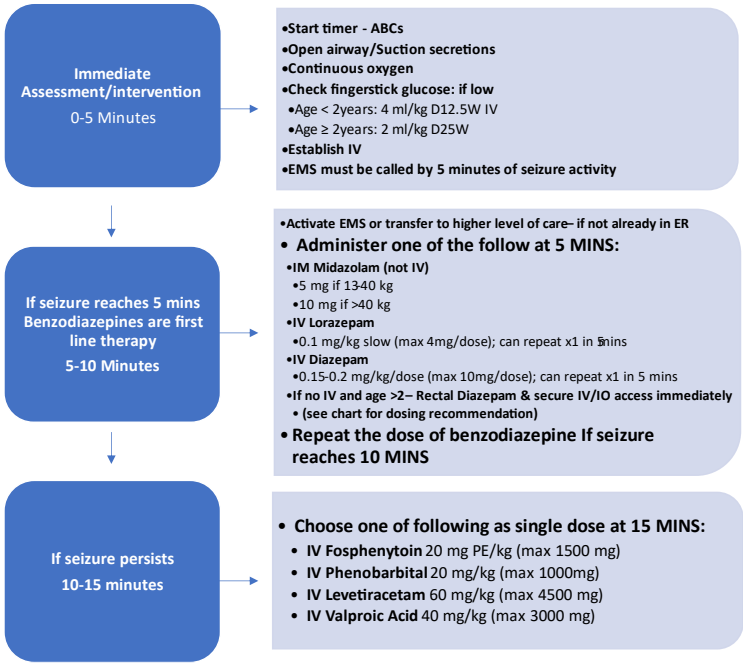
PEDIATRIC CARE

Pediatric Ambulatory Care Seizure Management



Pediatric Status Epilepticus Algorithm



Considerations

- For children less than 1m of age: activate EMS Status epilepticus is defined as convulsive seizure activity lasting 5 minutes, or multiple seizures without return to baseline
- Status epilepticus is the most common neurologic emergency of childhood with incident between 17 -23 per 100,000 per year [1]
- Children treated more aggressively and those with shorter episodes of status epilepticus are less likely to develop subsequent neurological deficits
- Home rescue medications for seizures > 5 minutes or >3 seizures in 1 hour

DIASTAT® AcuDial™ (diazepam rectal gel)
Dosing Recommendations (by age and weight)

2 to 5 years 0.5 mg/kg		6 to 11 years 0.3 mg/kg		12+ years 0.2 mg/kg	
Weight (kg)	Dose (mg)	Weight (kg)	Dose (mg)	Weight (kg)	Dose (mg)
6 to 10	5	10 to 16	5	14 to 25	5
11 to 15	7.5	17 to 25	7.5	26 to 37	7.5
16 to 20	10	26 to 33	10	38 to 50	10
21 to 25	12.5	34 to 41	12.5	51 to 62	12.5
26 to 30	15	42 to 50	15	63 to 75	15
31 to 35	17.5	51 to 58	17.5	76 to 87	17.5
26 to 44	20	59 to 74	20	88 to 111	20

Valtoco (Diazepam Nasal Spray)

6-11 years of age (0.3mg/kg) Weight (Kg)	12 years of age and older (0.2mg/kg) Weight (Kg)	Dose (mg)	Number of Nasal spray devices	Number of sprays
10-18 Kg	14 to 27	5	One 5 mg device	One spray in one nostril
19 to 37	28 to 50	10	One 10mg device	One spray in one nostril
38 to 55	51 to 75	15	Two 7.5mg devices	One spray in each nostril
56 to 74	76 and up	20	Two 10mg devices	One spray in each nostril

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Pediatric Seizure Guideline

Febrile Seizures

- Febrile seizures occur in children with fever, usually in the setting of systemic viral or bacterial infection. Affected patients are typically between the ages of 6 months and 6 years of age and do not have epilepsy, central nervous system infection or inflammation, or other triggers of seizures.
- Febrile seizures occur in 2 to 4 percent of children between ages 6 months and 6 years of age, with a peak incidence between 12 and 18 months. Febrile seizures are an age-dependent phenomenon, likely related to a vulnerability of the developing nervous system to the effects of fever in combination with an underlying genetic susceptibility. Aside from age, the most identified risk factors include high fever, viral infection, recent immunization, and a family history of febrile seizures.
- Most children have febrile seizures within 24 hours of fever onset, it may even be the first manifestation that the child is ill.
- Febrile seizures are a clinical diagnosis. In children with a typical history and a reassuring and non-focal exam, diagnostic testing is unnecessary in most cases. A genetic predisposition to febrile seizures has long been recognized, although the exact mode of inheritance is not known in most cases. Among first-degree relatives of children with febrile seizures, 10 to 20 percent of parents and siblings also have had or will have febrile seizures.
- The use of around the clock antipyretics does not prevent additional febrile seizures- That recommendation may increase the risk of toxicity and has no benefit.
- Lumbar puncture is unnecessary in most well-appearing children who have returned to a normal baseline after a febrile seizure. Post-ictal drowsiness typically resolved within five to ten minutes, depending upon the duration and type of seizure. Lumbar puncture should be performed when there are meningeal signs or symptoms or other clinical features that suggest possible meningitis or intracranial infection. Additional circumstances that warrant consideration of lumbar puncture include:
    - Infants between 6 and 12 months of age, if the immunization status for Hemophilus influenza type b or Streptococcus pneumonia is deficient or undetermined.
    - Current treatment with antibiotics since antibiotics can mask the signs and symptoms of meningitis.
    - Prolonged seizures, including febrile status epilepticus.
    - Seizures that occur after the second day of a febrile illness
- **Simple Febrile Seizure**
    - The most common and are characterized by seizures that last less than 15 minutes, have no focal features, and occur once in a 24-hour period.
    - These are mainly generalized tonic-clonic seizures but may also be atonic or tonic in character. Complex febrile seizures are characterized by episodes that last more than 15 minutes, have focal features or postictal paresis, or occur more than once in 24 hours.
    - No referral to child neurology necessary for simple febrile seizures.
- **Complex Febrile Seizure**
    - Seizures have focality (i.e. start with one limb jerking), last more than 15 minutes of recur within a 24-hour period.
    - Refer to Child Neurology for further evaluation.
    - Admit for observation if: more than 2 seizures in 24 hours or the patient has a prolonged post-ictal period.
    - Discharge if:
        - Cause of fever does not require inpatient admission.
        - Patient appears nontoxic and returns to neurologic baseline.
        - Parent education and parental anxiety addressed.

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First Non-Febrile Seizure

- Every year 25,000-40,000 Children experience their first non-febrile seizure.
- Treatment with anti-epilepsy medication is not indicated for the prevention of the development of epilepsy.
- Electroencephalogram (EEG) is useful in determining the type of seizure, risk for recurrence or epilepsy syndrome and should be performed after all first seizures and may be done as an outpatient procedure.
- Routine imaging is not necessary unless there is concern for an underlying process which may contribute to the new onset seizure.
    - If neuroimaging study is obtained, MRI is the preferred modality. CT may be performed in emergency situations.
- If age appropriate, consider toxicology screen.
- Refer to Child Neurology for EEG and further evaluation,

Seizure in known Epilepsy Patients

- Check anti-epileptic drug level.
- Refer to child neurology recommendations.
- Rule out underlying infection
- Ensure medication adherence.

Infantile Spasms

- Infantile Spasms is a unique age related (6-12m: although may occur outside that range) epilepsy characterized by epileptic spasms, neuro-developmental regression and EEG finding of hypsarrhythmia.
- Incidence of 2-3 per 10,000 live births
- Treatment options: ACTH, prednisolone, vigabatrin
- Urgent child neurology referral for further evaluation

References

1. Abend NS, Loddenkemper T. Pediatric status epilepticus management. Curr Opin Pediatr. 2014; 26:668-674
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3. "Treatment of Convulsive Status Epilepticus in Children and Adult" *Epilepsy Currents* 16.1 Jan/Feb 2016
4. Eric Silvermann et al. "Prehospital Care for the adult and pediatric Seizure Patient: Current Evidence-based Recommendations. Western Journal of Emergency Medicine, Volume 18
5. Evaluating a First Nonfebrile Seizure in children. American Academy of Neurology: Guideline. Current practice guideline. Reaffirmed on April 25, 2017